UNIT 5

The Network Layer

Complete Exam Notes with Diagrams, Q&A

Computer Networks

Comprehensive Study Material

May 2026

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5.1 Network Layer Design Issues

This section introduces the fundamental issues that network layer designers must address, including the service model provided to the transport layer and the internal organization of the network itself.

Core Responsibilities of the Network Layer

- Routing packets from source to destination across multiple hops
- Providing a service interface to the transport layer
- Managing the internal network organization
- Handling addressing, fragmentation, and congestion

5.1.1 Store-and-Forward Packet Switching

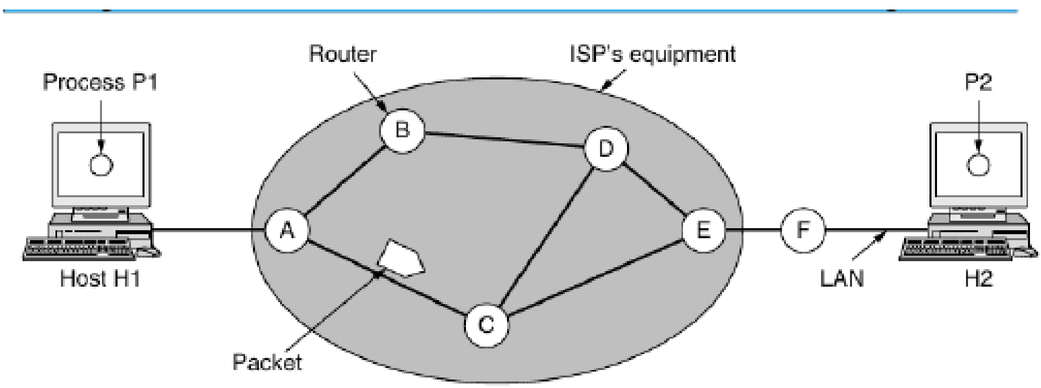


Figure 5-1. The environment of the network layer protocols.

Figure 5-1: *The environment of the network layer protocols. ISP equipment (shaded oval) consists of routers connected by transmission lines. Customer equipment (hosts and customer-owned routers) is outside the oval.*

The network layer protocols operate within an environment composed of two major components:

Table 5-1 Network Layer Environment Components

Component	Description
ISP's Equipment	Routers connected by transmission lines, shown inside the shaded oval.
Customers' Equipment	Hosts and customer-owned routers, shown outside the oval.

Example Topology:

- **Host H1** is directly connected to ISP router **A** (e.g., a home computer plugged into a DSL modem).
- **Host H2** resides on a LAN (e.g., an office Ethernet) with a customer-owned router **F**, which connects to the ISP via a leased line.
- **Router F** is shown outside the oval because it does not belong to the ISP. However, for routing algorithm analysis, customer-premise routers are considered part of the ISP network because they run the same routing algorithms.

Store-and-Forward Packet Switching Mechanism

1. A host with a packet to send transmits it to the nearest router (either on its own LAN or over a point-to-point link to the ISP).
2. The packet is **stored** at the router until it has fully arrived and the link has finished processing it (including checksum verification).
3. The packet is then **forwarded** to the next router along the path.
4. This process repeats until the packet reaches the destination host, where it is delivered.

Key Point

This is the classic **store-and-forward packet switching** mechanism. Each router must receive the entire packet before forwarding it, introducing a store-and-forward delay at each hop.

5.1.2 Services Provided to the Transport Layer

The network layer provides services to the transport layer at the **network layer/transport layer interface**. The design of these services must satisfy three critical goals:

Table 5-2 Network Layer Service Design Goals

Goal	Description
Router Technology Independence	The services should be independent of the underlying router technology.
Topology Shielding	The transport layer should be shielded from the number, type, and topology of the routers present.
Uniform Addressing	The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

The Fundamental Debate: Connection-Oriented vs. Connectionless Service

These goals give designers considerable freedom, which has historically led to a deep division between two competing philosophies:

Camp 1: The Internet Community (Connectionless)

- Philosophy: routers move packets, nothing else
- Premise: network is inherently unreliable
- Hosts must do error control and flow control
- Service: **connectionless** (SEND PACKET, RECEIVE PACKET)
- Each packet carries **full destination address**
- Exemplifies the **end-to-end argument**

Camp 2: Telephone Companies (Connection-Oriented)

- Philosophy: network should provide reliable service
- Premise: 100 years of telephone system success
- QoS is paramount for voice/video
- Service: **connection-oriented**
- Without connections, QoS is very difficult
- Requires setup before data transfer

Historical Evolution

Table 5-3 Historical Network Technologies

Era	Technology	Orientation	Status
1970s	X.25	Connection-oriented	Early data network
1980s	Frame Relay	Connection-oriented	Successor to X.25
ARPANET/Early Internet	IP	Connectionless	Grew tremendously
1980s-1990s	ATM	Connection-oriented	Now niche uses
Present	IP	Connectionless (dominant)	Taking over telephony

Important Nuance

Despite IP's connectionless dominance, the Internet is evolving connection-oriented features under the covers:

- **MPLS (MultiProtocol Label Switching):** Connection-oriented technology used within ISP networks.
- **VLANs:** Widely used (seen in Chapter 4).

5.1.3 Implementation of Connectionless Service (Datagram Networks)

Organization Principle

- Packets are injected into the network **individually** and routed **independently** of each other.
- **No advance setup** is required.
- Packets are called **datagrams** (by analogy with telegrams).
- The network is called a **datagram network**.

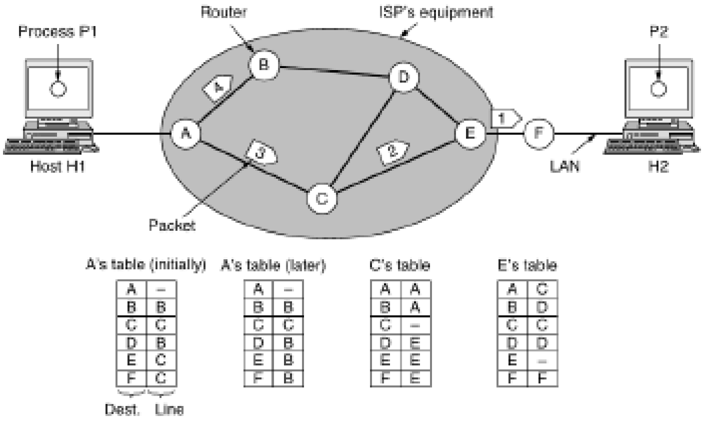
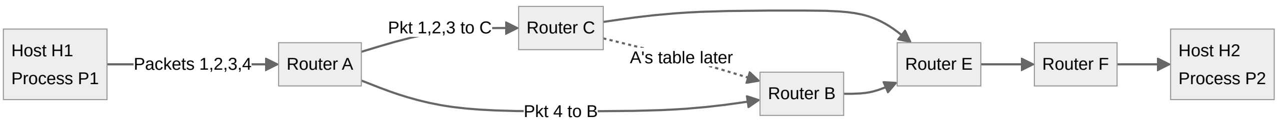


Figure 5-2. Routing within a datagram network.

Figure 5-2: *Routing within a datagram network. Router A's routing table maps destinations to outgoing lines. Packets are routed independently.*

How Datagram Routing Works: Detailed Example

Scenario: Process **P1** on host **H1** has a long message for process **P2** on host **H2**.



Process:

1. **Transport Layer (H1):** Prepends transport header, hands to network layer.
2. **Packet Fragmentation:** Message is four times max packet size, broken into packets **1, 2, 3, 4**.
3. **Router Operation:** Every router maintains a **routing table** mapping destinations to outgoing lines. Each entry: (Destination, Outgoing Line).

Table 5-4 Router A's Routing Tables

Destination	Line (Initially)	Line (Later)
A	--	--
B	B	B
C	C	C
D	B	B
E	C	B
F	C	B

Packet Journey:

- **Packets 1, 2, 3:** A → C → E → F → H2
- **Packet 4:** A → B (because A updated its routing table due to traffic jam along ACE path)
- The algorithm managing these tables is the **routing algorithm**.

IP: The Dominant Datagram Protocol

- **IP (Internet Protocol)** is the dominant connectionless network service.
- Each packet carries a **destination IP address** for individual forwarding.
- **IPv4:** 32-bit addresses; **IPv6:** 128-bit addresses.

5.1.4 Implementation of Connection-Oriented Service (Virtual-Circuit Networks)

Organization Principle

- A path must be **established before** any data packets can be sent.
- This connection is called a **VC (virtual circuit)**, by analogy with telephone circuits.
- The network is called a **virtual-circuit network**.
- Each packet carries an **identifier** telling which VC it belongs to.

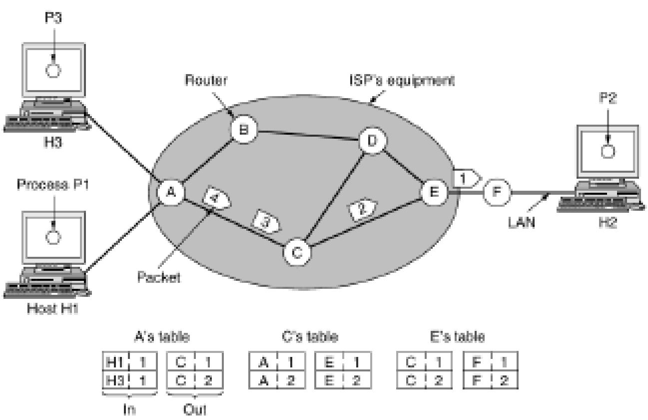


Figure 5-3. Routing within a virtual-circuit network.

Figure 5-3: *Routing within a virtual-circuit network. Router tables are indexed by connection identifiers. Routes are fixed at setup time.*

How Virtual-Circuit Routing Works: Detailed Example

Scenario: Host **H1** has established **connection 1** with host **H2**.

Connection 1: H1 → H2

Table 5-5 VC Routing Tables for Connection 1

Router	In	Out
A	From H1, connection ID 1	To C, connection ID 1
C	From A, connection ID 1	To E, connection ID 1
E	From C, connection ID 1	To F, connection ID 1

Second Connection: H3 → H2

- Host **H3** chooses **connection identifier 1** (as initiator, its only connection).
- This creates a **connection identifier conflict** at router C.

Table 5-6 Connection Identifier Conflict Resolution

Router	Can Distinguish?	Resolution
A	Yes - easily distinguishes packets from H1 vs. H3	None needed at input
C	No - cannot tell which connection a packet belongs to	A assigns a **different connection identifier** for the second connection

Label Switching

Routers must have the ability to **replace connection identifiers** in outgoing packets to avoid conflicts. This process is called **label switching**.

MPLS: A Connection-Oriented Network Service

- **MPLS (MultiProtocol Label Switching)** is a prominent connection-oriented service used within ISP networks.
- IP packets are wrapped in an MPLS header containing a **20-bit connection identifier (label)**.
- Often hidden from customers; ISPs establish long-term connections for large traffic volumes.

5.1.5 Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Figure 5-4. Comparison of datagram and virtual-circuit networks.

Figure 5-4: *Comparison of datagram and virtual-circuit networks across key dimensions.*

Table 5-7 Detailed Comparison: Datagram vs. Virtual-Circuit Networks

Issue	Datagram Network	Virtual-Circuit Network
Circuit setup	Not needed	Required
Addressing	Full source and destination address in each packet	Short VC number only
State information	Routers hold no connection state	Each VC requires router table space
Routing	Each packet routed independently	Route chosen at setup; all packets follow it
Router failure effect	Only packets lost during crash	All VCs through failed router terminated
Quality of service	Difficult	Easy if resources allocated in advance
Congestion control	Difficult	Easy if resources allocated in advance

Detailed Trade-off Analysis

1. Setup Time vs. Address Parsing Time

Aspect	Virtual Circuit	Datagram
Setup	Requires setup phase (takes time, consumes resources)	No setup needed
Forwarding	Easy: circuit number indexes table	More complicated lookup

2. Address Overhead

Datagram networks use **longer destination addresses** because addresses have global meaning. If packets are short, full destination address in every packet = significant overhead.

3. Router Table Space

Network Type	Table Requirement
Datagram	Entry for every possible destination
Virtual Circuit	Entry only for each virtual circuit

4. Quality of Service and Congestion Control

- **Virtual Circuits:** Resources can be **reserved in advance** during connection establishment. Clear advantage for QoS.
- **Datagrams:** No advance reservation; congestion avoidance is more difficult.

5. Suitability for Different Traffic Patterns

Traffic Type	Best Fit	Rationale
Short transactions (credit card verification)	Datagram	Setup/clear overhead dwarfs actual use
Long-running flows (VPN between offices)	Virtual Circuit (Permanent VC)	Permanent VCs lasting months/years useful

6. Fault Tolerance

Failure Scenario	Datagram	Virtual Circuit
Router crash	Only queued packets lost	All VCs through router must be aborted
Communication line loss	Route around failure easily	Fatal to all VCs using that line
Load balancing	Can change routes mid-transmission	Route fixed at setup time

Exam Tip: Key Takeaways for 5.1

1. The network layer provides either **connectionless (datagram)** or **connection-oriented (virtual circuit)** service.
2. **Store-and-forward packet switching** is the fundamental mechanism for packet traversal.
3. The Internet is predominantly connectionless (IP), but connection-oriented tech (MPLS, VLANs) is increasingly used for QoS.
4. The choice involves trade-offs in setup overhead, addressing, state management, fault tolerance, and QoS.

5.2 Routing Algorithms

Overview

The **primary function** of the network layer is routing packets from the source machine to the destination machine. In most networks, packets require **multiple hops** to complete their journey. The only notable exception is broadcast networks.

The **routing algorithm** is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.

Routing vs. Forwarding: Critical Distinction

Process	Description
Routing	Making the decision about which routes to use. **The planning phase.**
Forwarding	What happens when a packet actually arrives. Looking up outgoing line in routing tables and sending the packet.

A router has **two internal processes**:

1. **Forwarding process:** Handles each arriving packet, looks up the outgoing line, and transmits it.
2. **Routing algorithm process:** Fills in and updates the routing tables.

When Routing Decisions Are Made

Table 5-8 Timing of Routing Decisions

Network Type	When Decisions Are Made
Datagram networks	Anew for **every arriving data packet**
Virtual-circuit networks	Only when a **new VC is being set up**; data packets follow established route

Desirable Properties of Routing Algorithms

Table 5-9 Essential Properties of Routing Algorithms

Property	Description
Correctness	Must route packets to their intended destinations.
Simplicity	Should not be unnecessarily complex.
Robustness	Must cope with hardware/software failures and topology changes without network-wide reboots.
Stability	Must converge to fixed paths and stay there; converge quickly.
Fairness	Should treat connections equitably.
Efficiency	Should make effective use of network resources.

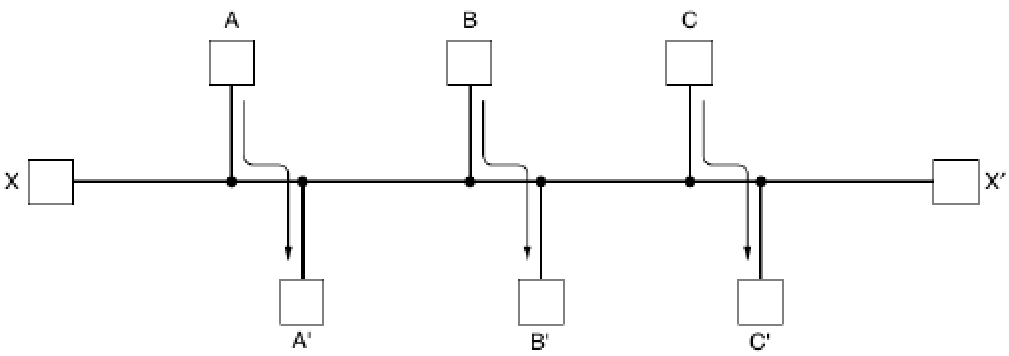


Figure 5-5. Network with a conflict between fairness and efficiency.

Figure 5-5: *Network with a conflict between fairness and efficiency. Traffic between A-A', B-B', C-C' saturates horizontal links.*

The Fairness-Efficiency Conflict

These goals are often contradictory:

- **Efficiency-maximizing:** Shut off all X-to-X' traffic to maximize total flow.
- **Fairness:** X and X' would not accept being completely denied service.
- **Conclusion:** A compromise between global efficiency and fairness is necessary.

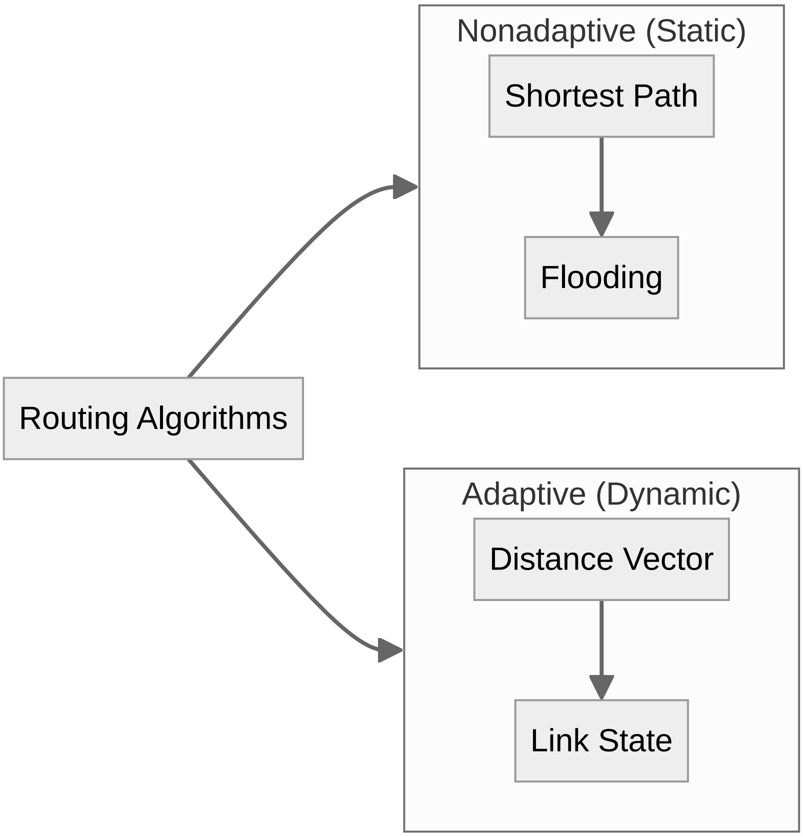
Classification of Routing Algorithms

1. Nonadaptive (Static) Routing

- Do not use measurements of current topology/traffic
- Routes computed **in advance, off-line**
- Downloaded when network boots
- Do **not respond to failures**
- Best for clear routing choices

2. Adaptive (Dynamic) Routing

- Change decisions to reflect topology/traffic changes
- Information from: local, adjacent, or all routers
- Timing: when topology changes or every ΔT seconds
- Metrics: distance, hops, estimated transit time



5.2.1 The Optimality Principle

> **Statement of the Principle (Bellman, 1957)**
>
> If router **J** is on the optimal path from router **I** to router **K**, then the optimal path from **J** to **K** also falls along the same route.

Proof by Contradiction: Let the optimal route from I to K be decomposed into r_1 (I to J) and r_2 (J to K). If a better route than r_2 existed from J to K, it could be concatenated with r_1 to produce a better route from I to K. This contradicts the assumption that r_1r_2 is optimal.

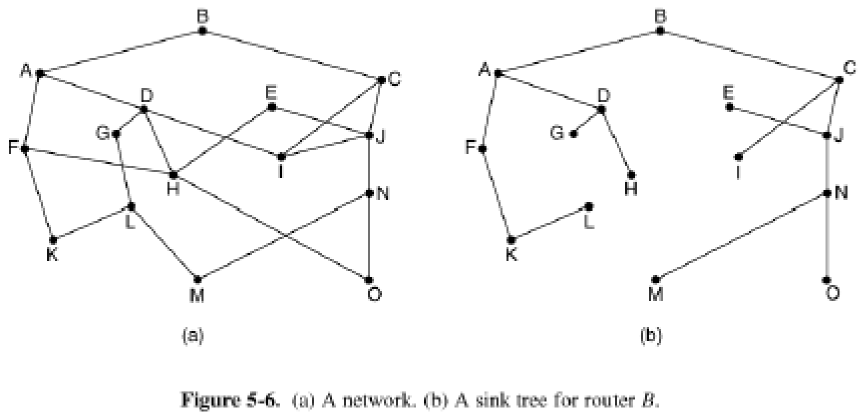


Figure 5-6: *(a) A general network topology. (b) The sink tree for router B - the set of optimal paths from all other routers to B.*

> **Consequence: The Sink Tree**
>
> The set of optimal routes from **all sources** to a **given destination** forms a **tree rooted at the destination**. This is called a **sink tree**.
>
> - **Tree structure:** Rooted at destination; no loops.
> - **Finite delivery:** Packet delivered within bounded number of hops.
> - **Generalization (DAG):** If all optimal paths are allowed, becomes a Directed Acyclic Graph.

> **Practical Considerations**
>
> - Links and routers go down and come back up dynamically.
> - Different routers may have different views of the topology.
> - Despite complications, the sink tree provides a **benchmark** for measuring routing algorithms.

5.2.2 Shortest Path Algorithm (Dijkstra's Algorithm)

Fundamental Approach

The shortest path algorithm computes optimal paths given a complete picture of the network. These are the paths that distributed routing algorithms aim to find.

Table 5-10 Graph-to-Network Mapping

Graph Element	Network Element
Node	Router
Edge	Communication line (link)

Metrics for Path Length

Table 5-11 Path Length Metrics

Metric	Description	Example
Number of hops	Count of edges traversed	Paths ABC and ABE equally long
Geographic distance	Physical distance in km	ABC much longer than ABE if drawn to scale
Mean delay	Measured delay of test packet	Fastest path, not fewest edges
General composite	Function of distance, bandwidth, traffic, cost	Computed by changing weighting function

Dijkstra's Algorithm (1959)

Goal: Find shortest paths from a **source** to **all destinations**.

Key Properties:

- Distances must be **non-negative**.
- Each node labeled with distance from source along best known path.

Label Types:

- **Tentative:** Initial state; may improve.
- **Permanent:** Represents shortest possible path; never changed.

Algorithm Steps:

1. Initialize all labels as **tentative**, set to **infinity**.
2. Make source node **permanent** with distance 0.
3. Examine neighbors of the newest permanent node; update tentative labels.
4. Select the tentative node with **smallest label**; make it permanent.
5. Repeat steps 3-4 until all nodes are permanent.

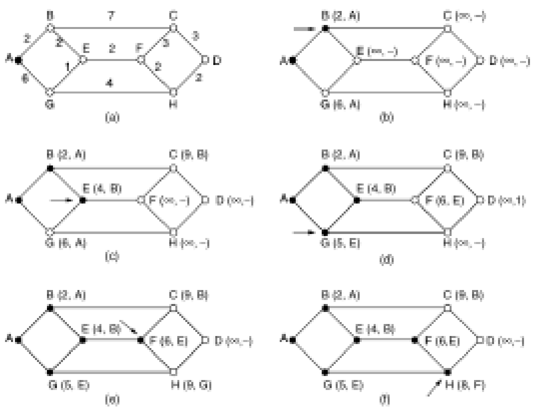


Figure 5-7. The first six steps used in computing the shortest path from *A* to *D*. The arrows indicate the working node.

Figure 5-7: *The first six steps used in computing the shortest path from A to D using Dijkstra's algorithm. Arrows indicate the working node.*

Network Topology for Figure 5-7

Table 5-12 Edge Weights for Dijkstra Example

Edge	Weight	Edge	Weight
A-B	2	A-E	2
A-G	6	B-C	7
B-E	2	E-F	2
E-G	1	G-H	4
C-F	3	C-D	3
F-H	2	H-D	2

Algorithm Execution Trace

Table 5-13 Dijkstra's Algorithm Step-by-Step

Step	Working Node	Action	Result
(b)	**A** (perm)	Examine B, E, G	B(2,A), E(2,A), G(6,A); C,D,F,H=∞
(c)	**B** (perm)	Examine C, E	C(9,B); E stays (2,A)
(d)	**E** (perm)	Examine F, G	F(4,E); G(3,E)
(e)	**F** (perm)	Examine C, H	C(7,F); H(6,E)
(f)	**G** (perm)	Examine H	H stays (6,E)

How to Answer Dijkstra's Algorithm Questions

1. Draw the network graph with all edge weights.
2. Create a table with columns for each node: (distance, predecessor).
3. Mark tentative distances as ∞ initially, except source = 0.
4. At each step: pick the unvisited node with smallest tentative distance, make it permanent.
5. Update neighbors: if new path is shorter, update their tentative distance.
6. Continue until destination is permanent or all nodes are processed.
7. Trace back predecessors to find the actual path.

5.2.4 Distance Vector Routing

Computer networks generally use **dynamic routing algorithms**. The two most popular are:

1. **Distance Vector Routing** (this section)
2. **Link State Routing** (next section - not detailed in notes)

Alternative Names: Distributed Bellman-Ford routing algorithm; **RIP (Routing Information Protocol)**.

Core Mechanism

Each router maintains a **routing table (vector)** containing:

- **Best known distance** to each destination
- **Preferred outgoing line** to reach that destination

Tables updated by **exchanging information with neighbors**.

Table Structure

Table 5-14 Distance Vector Routing Table

Field	Description
Destination	One entry for each router in network
Preferred outgoing line	Best next hop
Distance estimate	Estimated cost to reach destination

Update Process

Once every **T msec**, each router:

1. **Sends** to each neighbor a list of estimated delays to each destination.
2. **Receives** a similar list from each neighbor.
3. For neighbor **X** with delay estimate X_i to router i, and local delay **m** to X:
   Delay to i via X = X_i + m
4. Selects the **best estimate** and updates table.

> **Critical Point**
>
> The **old routing table is NOT used** in the calculation. Each update is computed entirely from neighbor vectors.

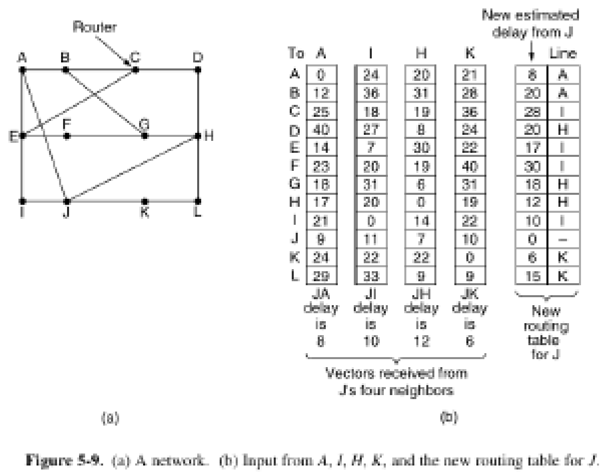


Figure 5-9: *(a) A network. (b) Input from A, I, H, K and the new routing table for J.*

Detailed Example: Router J

J's Delay to Neighbors: A = 8, I = 10, H = 12, K = 6

Computing New Route to Router G

Table 5-15 Distance Calculation to G via Each Neighbor

Via	Calculation	Total Delay
A	18 (A→G) + 8 (J→A)	26
I	31 (I→G) + 10 (J→I)	41
H	6 (H→G) + 12 (J→H)	**18** (Best)
K	31 (K→G) + 6 (J→K)	37

Result: J's routing table entry for G: **Delay = 18 msec, Route = via H**

The Count-to-Infinity Problem

Convergence Characteristics

News Type	Reaction Speed
Good news	Rapid - one vector exchange per hop
Bad news	Slow - converges leisurely (count-to-infinity)

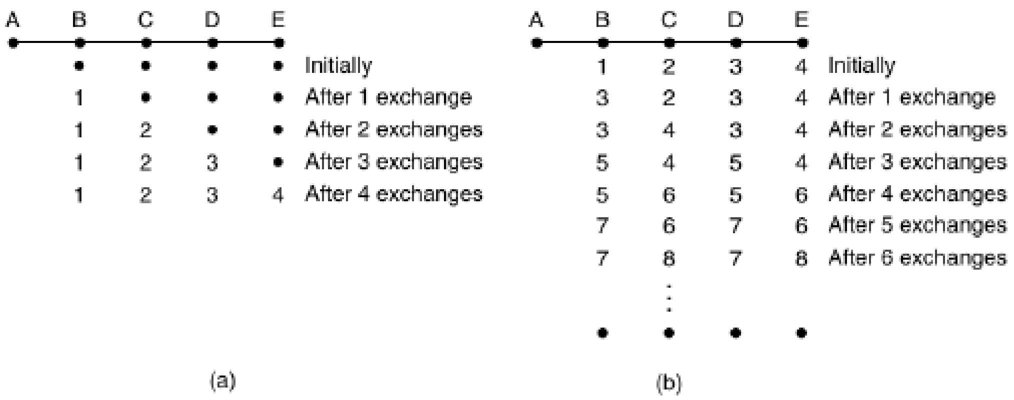


Figure 5-10. The count-to-infinity problem.

Figure 5-10: *(a) Good news propagation - spreads at one hop per exchange. (b) Bad news propagation - the count-to-infinity problem.*

Good News Propagation (Figure 5-10a)

Scenario: Five-node linear network (A-B-C-D-E), A initially down.

Table 5-16 Good News Propagation Speed

Exchange	B	C	D	E
Initially	∞	∞	∞	∞
After 1	1	∞	∞	∞
After 2	1	2	∞	∞
After 3	1	2	3	∞
After 4	1	2	3	4

Rate: Good news spreads at **one hop per exchange**. In a network with longest path of N hops, all routers know within **N exchanges**.

Bad News Propagation - The Problem (Figure 5-10b)

Scenario: All links up initially. A goes down.

Table 5-17 Count-to-Infinity Problem Trace

Exchange	B	C	D	E	Explanation
Initially	1	2	3	4	Normal state
After 1	3	2	3	4	B doesn't hear from A; C says path=2, B thinks via C = 3
After 2	3	4	3	4	C sees neighbors claim 3, updates to 4
After 3	5	4	5	4	D updates to 5
...	...	...	...	...	Gradually increases toward infinity

Why Bad News Travels Slowly

- No router ever has a value more than **one higher** than the minimum of all neighbors.
- Gradually all routers work up to infinity.
- **Core Problem:** When X tells Y it has a path somewhere, **Y has no way of knowing if it itself is on that path**.
- **Recommendation:** Set infinity to **longest path + 1**.

Attempted Solutions

Table 5-18 Solutions to Count-to-Infinity

Technique	Description	Effectiveness
Split horizon with poisoned reverse	Prevent routers from advertising best paths back to neighbors they heard them from (RFC 1058)	Does not work well in practice

Fundamental Issue

Despite many heuristic attempts, **none work well in practice** because the fundamental information asymmetry remains: a router cannot determine if it lies on a neighbor's advertised path.

Exam Tip: Key Takeaways for 5.2

1. **Routing** = decision-making; **Forwarding** = packet handling.
2. Algorithms must balance correctness, simplicity, robustness, stability, fairness, and efficiency.
3. **Nonadaptive (static)** algorithms compute routes in advance; **Adaptive (dynamic)** respond to changes.
4. **Optimality principle:** optimal routes form a **sink tree**.
5. **Dijkstra's algorithm:** computes shortest paths using non-negative weights with tentative/permanent labels.
6. **Distance vector routing:** routers exchange delay vectors with neighbors.
7. **Count-to-infinity:** good news fast (1 hop/exchange), bad news slow.
8. **Split horizon with poisoned reverse** attempts to mitigate but doesn't solve the core problem.

5.6 The Network Layer in the Internet

5.6.0 Design Principles of the Internet

The foundational principles driving the Internet's design are enumerated in **RFC 1958**, drawing from Clark (1988) and Saltzer et al. (1984).

Table 5-19 Top 10 Internet Design Principles

Rank	Principle	Explanation
1	**Make sure it works.**	Multiple prototypes before finalizing standards.
2	**Keep it simple.**	Use simplest solution; fight features (Occam's razor).
3	**Make clear choices.**	If several ways exist, choose **one**.
4	**Exploit modularity.**	Protocol stacks with independent layers.
5	**Expect heterogeneity.**	Different hardware, facilities, applications.
6	**Avoid static options.**	Have sender and receiver negotiate values.
7	**Good design > perfect.**	Go with good design; burden edge cases.
8	**Strict sending, tolerant receiving.**	Send compliant packets; accept non-conformant.
9	**Think scalability.**	No centralized databases; spread load evenly.
10	**Consider performance and cost.**	Poor performance = nobody uses it.

Internet Architecture Overview

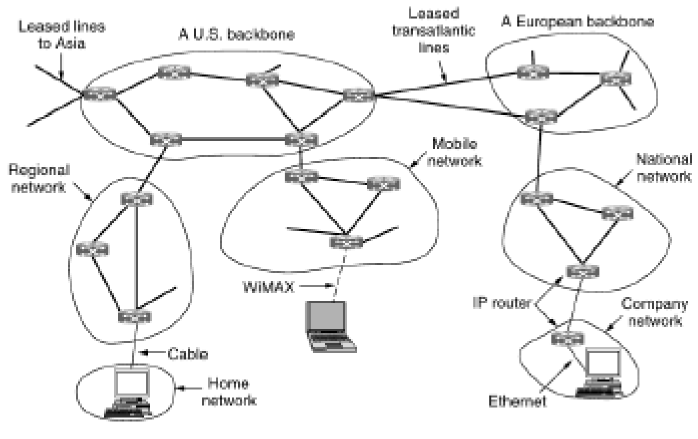


Figure 5-45. The Internet is an interconnected collection of many networks.

Figure 5-45: *The Internet as an interconnected collection of many networks: backbones, regional networks, ISPs, home networks, mobile networks.*

Table 5-20 Internet Components

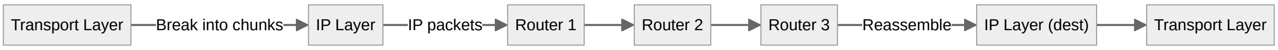
Component	Description
Tier 1 networks	Biggest backbones; high-bandwidth lines and fast routers.
Backbones	High-bandwidth infrastructure (U.S., European, etc.)
ISPs	Provide Internet access to homes and businesses.
Data centers	Full of server machines serving content.
Regional networks	Connect to backbones and smaller ISPs.
Edge networks	LANs at universities, companies, home networks, mobile.

Typical Packet Journey

A packet from a home network may traverse **four networks** and numerous IP routers before reaching a destination on a company network. Redundant connectivity exists throughout, creating many possible paths.

The IP Protocol's Core Function

The network layer provides a **best-effort** (i.e., **not guaranteed**) way to transport packets from source to destination.



IP Packet Size

- **In theory:** Up to **64 KB** each.
- **In practice:** Usually not more than **1500 bytes** (to fit in one Ethernet frame).

5.6.1 The IP Version 4 Protocol

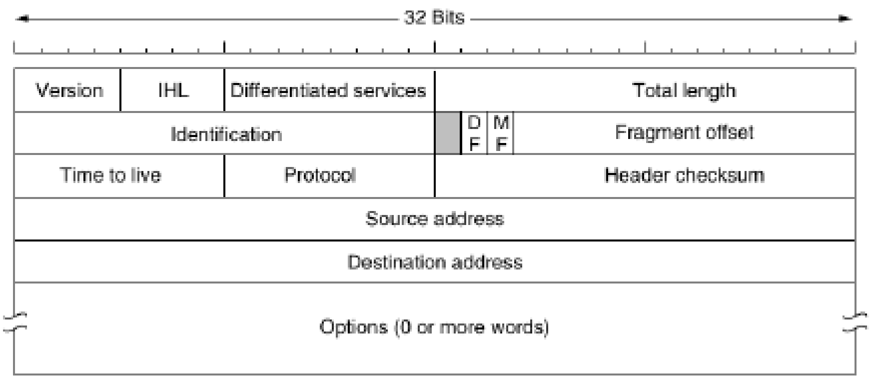


Figure 5-46. The IPv4 (Internet Protocol) header.

Figure 5-46: *The IPv4 (Internet Protocol) header format. 32 bits wide, transmitted left-to-right, top-to-bottom, big-endian.*

An IPv4 datagram consists of:

- **Header:** 20-byte fixed part + variable-length optional part
- **Body/Payload:** The actual data being transported

Field-by-Field Analysis of IPv4 Header

Table 5-21 IPv4 Header Fields

Field	Size	Description
Version	4 bits	4 = IPv4, 6 = IPv6. Allows version transition.
IHL	4 bits	Internet Header Length in 32-bit words. Min=5 (20 bytes), Max=15 (60 bytes).
Differentiated Services	8 bits	Originally Type of Service (TOS). Top 6 bits = service class; Bottom 2 bits = ECN.
Total Length	16 bits	Header + data. Maximum 65,535 bytes.
Identification	16 bits	Allows destination to determine which fragment a packet belongs to.
Flags (DF, MF)	3 bits	DF = Don't Fragment; MF = More Fragments.
Fragment Offset	13 bits	Position of fragment in original packet. Unit = 8 bytes.
Time to Live (TTL)	8 bits	Counts hops (decremented each hop). Packet discarded at 0. Prevents infinite loops.
Protocol	8 bits	Which transport process to give packet to (TCP=6, UDP=17).
Header Checksum	16 bits	One's complement sum of 16-bit halfwords. Must be recomputed at each hop.
Source Address	32 bits	IP address of source network interface.
Destination Address	32 bits	IP address of destination network interface.
Options	Variable	Variable length, padded to multiple of 4 bytes.

Fragmentation Fields Working Together

Identification + MF + Fragment offset implement fragmentation:

- All fragments of a packet contain the same **Identification** value.
- **MF** set on all fragments except the last one.
- **Fragment offset** tells where the fragment belongs in the original packet.
- Elementary fragment unit = **8 bytes**.

IP Options

Table 5-22 IPv4 Options

Option	Description	Status
Security	How secret the information is.	Ignored by all routers.
Strict source routing	Complete path as sequence of IP addresses.	Useful for emergency/timing.
Loose source routing	Must traverse listed routers, may pass through others.	For forcing particular paths.
Record route	Each router appends its IP address.	40 bytes too small now.
Timestamp	Like record route plus 32-bit timestamp.	Network measurement.

Current Status of IP Options

IP options have **fallen out of favor**. Many routers ignore them or don't process them efficiently. They are **only partly supported and rarely used**.

5.6.2 IP Addresses

Fundamental Characteristics

Table 5-23 IP Address Fundamentals

Aspect	Detail
Size	32-bit addresses
Scope	Every host and router has an IP address
Usage	Source and Destination address fields of IP packets
Important	IP address refers to a **network interface**, not a host

Multi-homed Hosts and Routers

- Host on one network: **One IP address** (typical)
- Host on two networks: **Two IP addresses** (one per interface)
- Routers: **Multiple IP addresses** (one per interface)

Prefixes and Hierarchical Structure

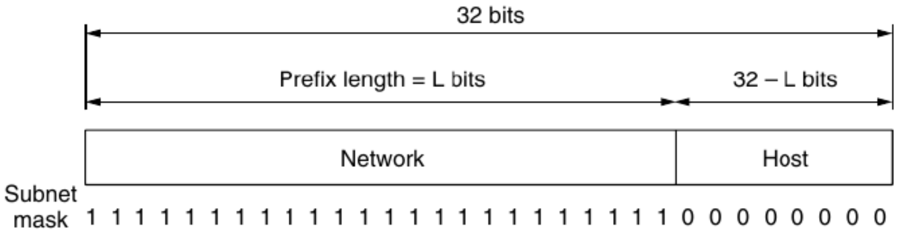


Figure 5-48. An IP prefix and a subnet mask.

Figure 5-48: *An IP prefix and a subnet mask. The prefix length = L bits for network, 32-L bits for host.*

Each 32-bit address consists of:

Portion	Location	Description
Network portion	Top bits (variable length)	Same for all hosts on a single network
Host portion	Bottom bits (remaining)	Unique within the network

> ### Prefix Notation (CIDR)
>
> A prefix is written as: `lowest_IP_address/prefix_length`
>
> **Example:** `128.208.0.0/24` = 2^8 = 256 addresses (24 network bits, 8 host bits)

> ### Subnet Mask
>
> Binary mask of 1s in the network portion. ANDed with IP address to extract network portion.
>
> **Example for /24:** `255.255.255.0` = 11111111 11111111 11111111 00000000

Advantages and Disadvantages of Hierarchical Addresses

Table 5-24 Hierarchical Addressing Pros and Cons

Aspect	Detail
Key advantage	Routers forward based only on **network portion**. Routing tables: ~300,000 prefixes vs. ~1 billion hosts.
Disadvantage 1	**Location-dependent:** IP address depends on network location. Mobile IP needed for moving hosts.
Disadvantage 2	**Address waste:** Hierarchy is wasteful unless carefully managed.

Subnets

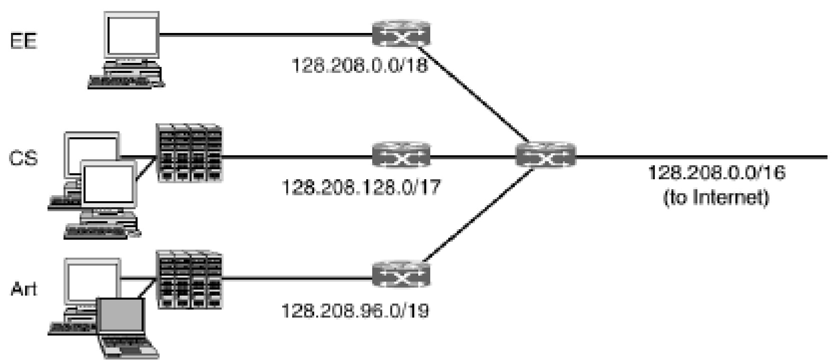


Figure 5-49. Splitting an IP prefix into separate networks with subnetting.

Figure 5-49: *Splitting an IP prefix into separate networks with subnetting: CS (/17), EE (/18), Art (/19).*

Problem: University has /16 prefix (65,536 addresses) for CS. Later EE and Art want connectivity.

> **Subnetting**
>
> - **Subnetting:** Splitting a block of addresses into several parts for internal use while acting like a single network externally.
> - **Subnets:** The resulting networks from dividing a larger network.

Table 5-25 Subnetting Example for 128.208.0.0/16

Department	Prefix	Size	Fraction
Computer Science	128.208.128.0/17	32,768	1/2
Electrical Eng.	128.208.0.0/18	16,384	1/4
Art	128.208.96.0/19	8,192	1/8
Unallocated	--	8,192	1/8

Router Forwarding with Subnets

When a packet arrives at the main router:

1. AND the destination address with each subnet mask
2. Check if result matches the corresponding prefix

Example: Packet for `128.208.2.151`

- AND with CS mask (255.255.128.0): `128.208.0.0` ≠ `128.208.128.0` → **No match**
- AND with EE mask (255.255.192.0): `128.208.0.0` = `128.208.0.0` → **Match!** → Forward to EE

> **Key Point**
>
> Subnet divisions can be changed later by updating subnet masks at internal routers. No need to contact ICANN or change external databases.

CIDR - Classless InterDomain Routing

> **Problem: Routing Table Explosion**
>
> - **Edge routers:** Can use default route to ISP for external destinations.
> - **Core routers:** In the **default-free zone** - must know routes to every network. Tables potentially millions of entries.

Solution: Route Aggregation

- **Route aggregation:** Combining multiple small prefixes into one larger prefix.
- **Supernet:** The resulting larger prefix (opposite of subnet).
- **CIDR:** Works with **variable-length prefixes**.

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

Figure 5-50. A set of IP address assignments.

Figure 5-50: *A set of IP address assignments for a block of 8,192 addresses (/19).*

Table 5-26 University Address Block Example (194.24.0.0/19)

University	Needed	Assigned Range	Mask	Prefix
Cambridge	2,048	194.24.0.0 - 194.24.7.255	255.255.248.0	194.24.0.0/21
Edinburgh	1,024	194.24.8.0 - 194.24.11.255	255.255.252.0	194.24.8.0/22
(Available)	1,024	194.24.12.0 - 194.24.15.255	--	194.24.12.0/22
Oxford	4,096	194.24.16.0 - 194.24.31.255	255.255.240.0	194.24.16.0/20

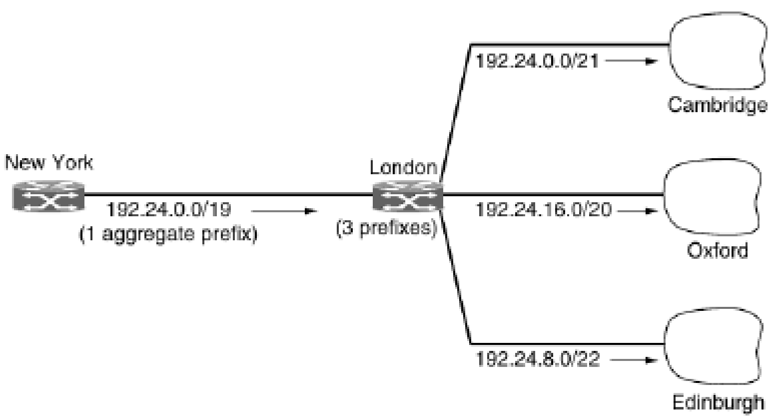


Figure 5-51. Aggregation of IP prefixes.

Figure 5-51: *Aggregation of IP prefixes: London router has 3 entries; New York router receives 1 aggregate prefix (194.24.0.0/19).*

How Aggregation Works

- **London router:** Needs entries for all three university prefixes (different outgoing lines).
- **New York router:** Receives **one aggregate prefix** 194.24.0.0/19 from London (covers all 8K addresses).
- **Result:** Three prefixes reduced to one, reducing New York's routing table.

Longest Matching Prefix Routing

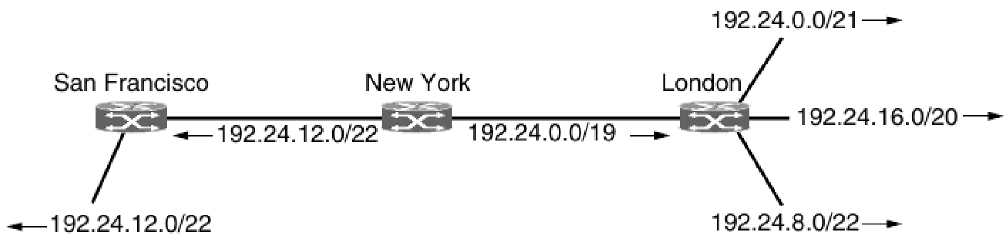


Figure 5-52. Longest matching prefix routing at the New York router.

Figure 5-52: *Longest matching prefix routing at the New York router. The /22 to San Francisco is more specific than the /19 to London.*

Longest Matching Prefix Rule

Packets are sent in the direction of the most specific route - the longest matching prefix (fewest IP addresses).

Example: 194.24.12.0/22 allocated to San Francisco within 194.24.0.0/19:

Prefix	Destination
194.24.0.0/19	London (aggregate)
194.24.12.0/22	San Francisco (more specific)

At New York router: addresses in 194.24.12.0/22 → **San Francisco** (/22 > /19).

Classful Addressing (Historical, Pre-1993)

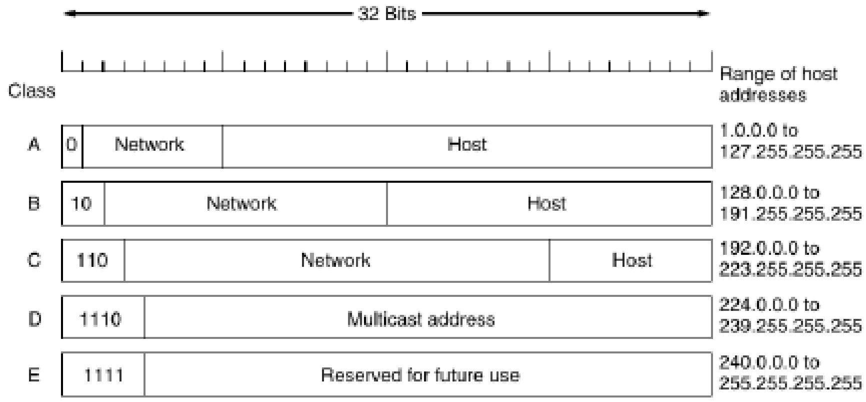


Figure 5-53. IP address formats.

Figure 5-53: *IP address formats for Class A, B, C, D, and E networks.*

Table 5-27 Classful Addressing

Class	Leading Bits	Network Bits	Host Bits	Address Range	Hosts/Network
A	0	7	24	1.0.0.0 - 127.255.255.255	16,777,216
B	10	14	16	128.0.0.0 - 191.255.255.255	65,536
C	110	21	8	192.0.0.0 - 223.255.255.255	256
D	1110	--	--	224.0.0.0 - 239.255.255.255	Multicast
E	1111	--	--	240.0.0.0 - 255.255.255.255	Reserved

The "Three Bears" Problem (Class B Crisis)

- **Class A** (16M hosts): Too large for most organizations.
- **Class B** (65K hosts): "Just right" but actually far too large for most.
- **Class C** (256 hosts): Too small.
- **Reality:** More than half of Class B networks had **fewer than 50 hosts**.
- **Waste:** Class C would have sufficed, but organizations feared future growth.

Transition to CIDR

- Subnets introduced for flexible internal assignment.
- CIDR added to reduce global routing table size.
- **Today:** Class bits are no longer used, though references remain common.

Special IP Addresses

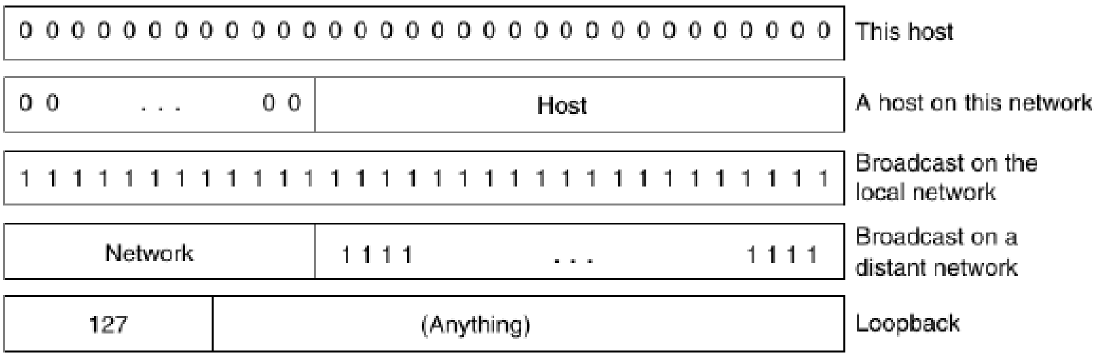


Figure 5-54. Special IP addresses.

Figure 5-54: *Special IP addresses: this host, broadcast, loopback, and network-specific addresses.*

Table 5-28 Special IP Addresses

Address	Binary Form	Meaning
0.0.0.0	00000000...	"This host" - used during booting
0.0.x.x	Network = 0	"A host on this network"
255.255.255.255	All 1s	Broadcast on the **local network**
network.255...	Host = all 1s	Broadcast on a **distant network**
127.x.x.x	01111111...	**Loopback** - processed locally

NAT - Network Address Translation

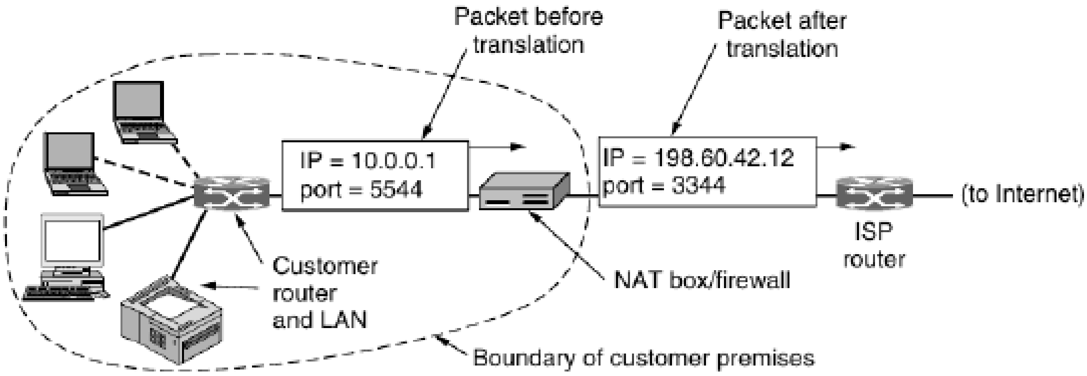


Figure 5-55. Placement and operation of a NAT box.

Figure 5-55: *Placement and operation of a NAT box. Internal addresses (10.x.y.z) are translated to a public IP (198.60.42.12).*

The IP Address Scarcity Problem

- ISP with /16: 65,534 usable host numbers.
- More customers than addresses - running out is happening **now**.

NAT: The Quick Fix (RFC 3022)

- **ISP assignment:** Single IP address per home/business for Internet traffic.
- **Internal addressing:** Every computer gets a unique **private IP address**.
- **Translation point:** Just before exiting to ISP, internal address → shared public address.

Private Address Ranges (RFC 1918)

Table 5-29 Private IP Address Ranges

Range	Prefix	Hosts Available
10.0.0.0 - 10.255.255.255	/8	16,777,216
172.16.0.0 - 172.31.255.255	/12	1,048,576
192.168.0.0 - 192.168.255.255	/16	65,536

Rule

No packets containing these private addresses may appear on the public Internet.

How NAT Works: TCP/UDP Port Mapping

Table 5-30 NAT Translation Process

Direction	Steps
Outgoing packet	1. Replace 10.x.y.z source with public IP 2. Replace TCP Source port with index into NAT's 65,536-entry table 3. Table entry contains: original IP + original port 4. Recompute checksums
Incoming packet	1. Extract Source port from TCP header 2. Use as index into mapping table 3. Retrieve internal IP and original port 4. Insert into packet; recompute checksums 5. Forward for normal delivery

Why Replace the Source Port?

Connections from 10.0.0.1 and 10.0.0.2 may both use port 5000. Source port alone is insufficient to identify the sending process. **Analogy:** Company with one telephone number; callers reach an operator who asks for an extension.

Objections to NAT

Table 5-31 Seven Objections to NAT

#	Objection	Explanation
1	**Violates IP architecture**	IP says each address uniquely identifies one machine. NAT: thousands use 10.0.0.1.
2	**Breaks end-to-end connectivity**	NAT mappings set up by outgoing packets; incoming cannot be accepted until after outgoing.
3	**Makes Internet connection-oriented**	NAT box must maintain mapping state. If it crashes, all TCP connections destroyed.
4	**Violates protocol layering**	NAT inspects TCP/UDP headers (layer 4) from layer 3. If TCP upgraded, NAT fails.
5	**Breaks non-TCP/UDP protocols**	New transport protocols fail because NAT cannot locate TCP Source port.
6	**Breaks embedded addresses**	FTP inserts IP addresses in body; NAT knows nothing about these.
7	**Limited by port space**	16-bit port = at most ~61,440 machines per IP (first 4,096 reserved).

Why NAT Persists

- IP address shortage - only expedient technique during IPv6 transition
- Firewall integration - blocks unsolicited incoming packets
- Privacy - hides internal network structure

5.6.3 IP Version 6

Background and Motivation

IPv4's exponential growth success created a critical problem: **IPv4 is close to running out of addresses**.

Table 5-32 Why IPv6 Was Needed

Issue	Detail
Address exhaustion	Even with CIDR and NAT, last IPv4 addresses expected to be assigned before end of 2012
Smartphone proliferation	Millions of mobile devices requiring constant connectivity
Industry convergence	Computer, communication, entertainment industries merging
Multimedia demand	Billions of machines for audio/video on demand

IETF Goals for IPv6

Table 5-33 IPv6 Design Goals

#	Goal	Description
1	**Scale**	Support billions of hosts
2	**Routing efficiency**	Reduce routing table sizes
3	**Simplicity**	Faster router processing
4	**Security**	Better authentication and privacy
5	**QoS**	Real-time data support
6	**Multicasting**	Specify scopes
7	**Mobility**	Host roaming without address change
8	**Evolvability**	Allow future evolution
9	**Coexistence**	Old and new protocols coexist for years

Major Improvements of IPv6

Table 5-34 IPv6 Improvements over IPv4

#	Improvement	Description
1	**Longer addresses**	128-bit addresses - effectively unlimited supply
2	**Simplified header**	Only 7 fields (vs. 13 in IPv4) - faster processing
3	**Better option support**	Previously required fields now optional
4	**Security**	Authentication and privacy as key features
5	**Quality of service**	Greater urgency due to multimedia growth

The IPv6 Fixed Header

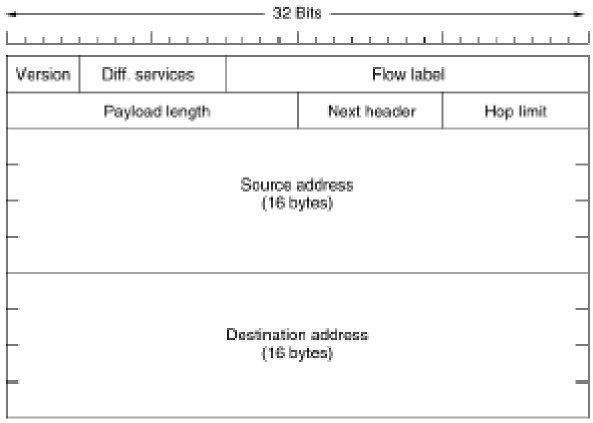


Figure 5-56. The IPv6 fixed header (required).

Figure 5-56: *The IPv6 fixed header (required). Total size: 40 bytes. Version, Diff. services, Flow label, Payload length, Next header, Hop limit, Source (128 bits), Destination (128 bits).*

Table 5-35 IPv6 Header Fields

Field	Size	Description
Version	4 bits	6 = IPv6, 4 = IPv4
Differentiated Services	8 bits	Distinguishes class of service (same as IPv4 DS + ECN)
Flow Label	20 bits	Marks groups of packets needing identical treatment (pseudo-connection)
Payload Length	16 bits	Bytes following 40-byte header (max 65,535 bytes)
Next Header	8 bits	Identifies extension header or transport protocol (TCP/UDP)
Hop Limit	8 bits	Decremented each hop; packet discarded at 0 (same as TTL)
Source Address	128 bits	16 bytes
Destination Address	128 bits	16 bytes

IPv6 Fixed Header: 40 bytes total

The IPv6 header is **twice as large** as IPv4's minimum (20 bytes) but requires **less processing** because options are handled via extension headers, and the header length is fixed.

IPv6 Address Notation

Standard format: `8000:0000:0000:0000:0123:4567:89AB:CDEF`

Table 5-36 IPv6 Address Notation Rules

Rule	Example
1. Omit leading zeros	0123 → 123
2. Replace 0000 groups with ::	8000:0000:0000:0000:0123... → 8000::123:4567:89AB:CDEF
3. IPv4 as :: + dotted decimal	::192.31.20.46

Address Space Magnitude

- Total addresses: $\mathbf{2^{128} \approx 3 \times 10^{38}}$
- If entire Earth covered with computers: **7×10^{23} addresses per m^2**
- This exceeds **Avogadro's number** (6.022×10^{23})

What Was Removed from IPv4

Table 5-37 IPv4 Fields Removed in IPv6

IPv4 Field	IPv6 Status	Rationale
IHL	Removed	IPv6 header has fixed length (40 bytes)
Protocol	Replaced by Next header	Tells what follows last IP header
Fragmentation fields	Removed	Different approach to fragmentation
Header checksum	Removed	Performance penalty not worth value; other layers have checksums

IPv6 Fragmentation

Table 5-38 IPv6 Fragmentation Approach

Aspect	Detail
Host responsibility	All hosts determine packet size using **path MTU discovery**
Router behavior	If packet too large, router **drops it** and sends error to source
Minimum forwardable	Raised from 576 to **1280 bytes**
Key difference	**Only source host can fragment**; routers may NOT fragment

Extension Headers

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

Figure 5-57. IPv6 extension headers.

Figure 5-57: *IPv6 extension headers: hop-by-hop, destination, routing, fragmentation, authentication, encrypted security payload.*

Table 5-39 IPv6 Extension Headers

Extension Header	Description
Hop-by-hop options	Miscellaneous info for routers
Destination options	Additional info for destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of sender's identity
Encrypted security payload	Info about encrypted contents

Extension Header Properties

- Each is **optional**.
- If multiple present, must appear in listed order after fixed header.
- Variable-length option encoding: **(Type, Length, Value)** tuple.
- Type field first 2 bits tell routers what to do if they don't understand the option:
  - 00 = Skip; 01 = Discard; 10 = Discard + ICMP error; 11 = Discard, no ICMP for multicast

Hop-by-Hop Header (Figure 5-58)

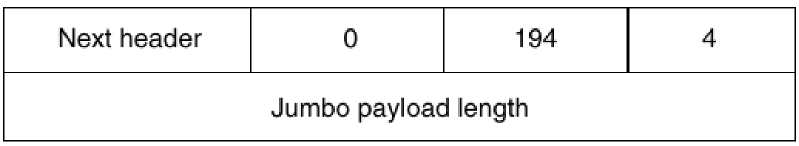


Figure 5-58. The hop-by-hop extension header for large datagrams (jumbograms).

Figure 5-58: *The hop-by-hop extension header for large datagrams (jumbograms). Option code 194 defines datagram size.*

- **Purpose:** Information that **all routers must examine**.
- **Jumbograms:** Datagrams exceeding 64 KB using option code 194.
- **Critical for supercomputers** transferring gigabytes.

Routing Header (Figure 5-59)

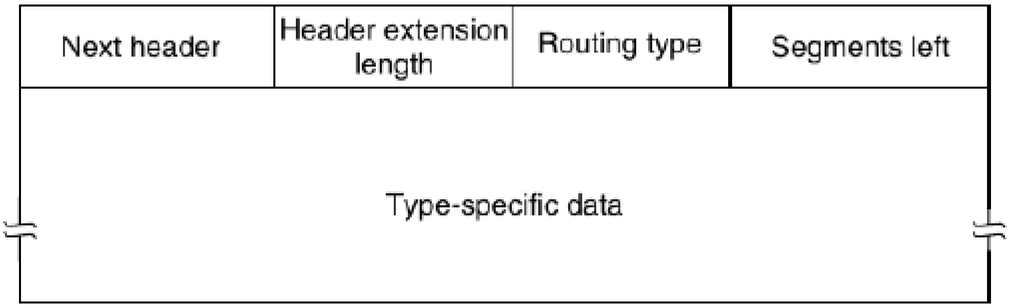


Figure 5-59. The extension header for routing.

Figure 5-59: *The extension header for routing. Lists routers that must be visited en route to destination.*

- **Function:** Lists routers that **must be visited** en route.
- **IPv4 equivalent:** Loose source routing.
- **Segments left field:** Tracks unvisited addresses; decremented at each visit.

IPv6 Deployment Reality

Table 5-40 IPv6 Deployment Status

Aspect	Status
Transition plan	Isolated IPv6 islands communicating via tunnels; grow and merge
Actual deployment	**Achilles heel** - remains little used despite full OS support
Primary use cases	New situations needing large numbers of addresses (mobile operators)
Transition strategies	Automatic tunnel config; dual-stack hosts (IPv4 + IPv6)
Future	Substantial deployment inevitable when IPv4 addresses exhausted

5.6.4 Internet Control Protocols

In addition to IP for data transfer, the Internet has several **control protocols** in the network layer:

Table 5-41 Internet Control Protocols

Protocol	IPv4	IPv6	Function
ICMP	ICMPv4	ICMPv6	Error reporting and network testing
ARP	ARP	NDP	Address resolution
DHCP	DHCPv4	DHCPv6	Host configuration

ICMP - The Internet Control Message Protocol

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

Figure 5-60. The principal ICMP message types.

Figure 5-60: *The principal ICMP message types: destination unreachable, time exceeded, echo reply, etc.*

Purpose: When something unexpected occurs during packet processing, the event is reported to the sender via ICMP. Each ICMP message type is carried **encapsulated in an IP packet**.

Table 5-42 Principal ICMP Message Types

Message Type	Description	Key Details
Destination unreachable	Packet could not be delivered	Router cannot locate destination; or DF bit packet too big
Time exceeded	TTL field hit 0	Packet looping or counter too low
Parameter problem	Invalid header field	Bug in sending host's IP software
Source quench	Choke packet	**Rarely used now**; congestion at transport layer
Redirect	Teach router about geography	Tells host to update to better route
Echo / Echo reply	Check if machine is alive	Used in **ping** utility
Timestamp request/reply	Same as Echo + timestamp	Network performance measurement
Router advertisement/solicitation	Find a nearby router	Hosts learn router IP addresses

TTL and Traceroute

Traceroute (Van Jacobson, 1987)

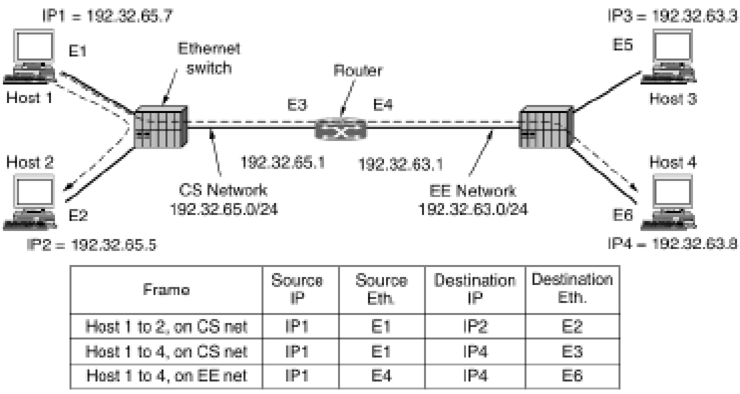
- Send sequence of packets with **TTL = 1, 2, 3, ...** successively.
- Counters reach zero at successive routers; each sends **TIME EXCEEDED** back.
- Result: IP addresses of all routers along the path, with timings.
- Perhaps the most useful network debugging tool of all time.

Source Quench (Deprecated)

Deprecated

Originally used to throttle hosts sending too many packets. Problem: during congestion, these packets add more fuel to the fire. Now rarely used; congestion control handled in **transport layer**.

ARP - The Address Resolution Protocol



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

Figure 5-61. Two switched Ethernet LANs joined by a router.

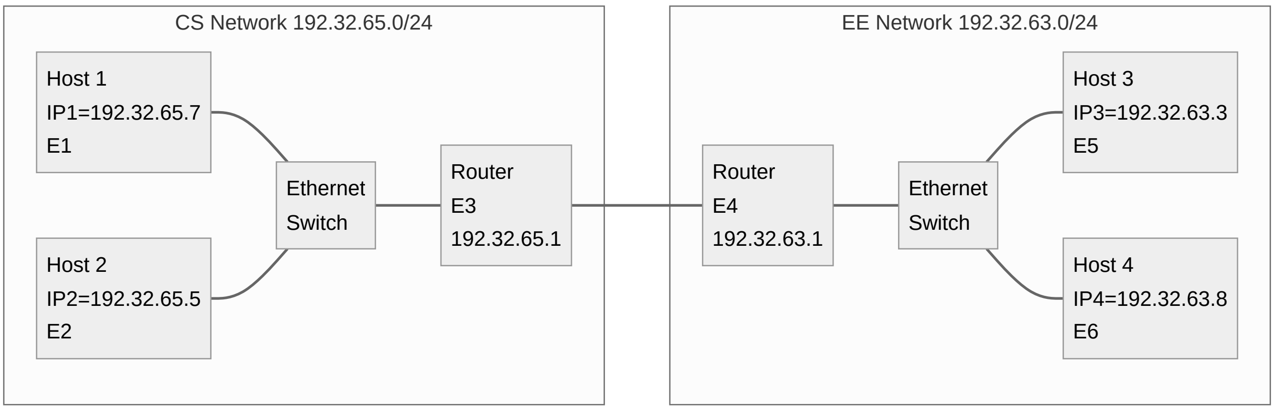
Figure 5-61: *Two switched Ethernet LANs joined by a router. IP addresses remain constant; Ethernet addresses change each hop.*

The Core Problem

Table 5-43 Layer Addressing Mismatch

Layer	Address Type	Size	Understanding
Network (IP)	IP address	32 bits	Routers and hosts understand
Data link (Ethernet)	Ethernet address	48 bits	NICs understand; unique per device

Ethernet NICs know nothing about 32-bit IP addresses. ARP bridges this gap.



ARP Operation: Intra-Network (Host 1 → Host 2)

Table 5-44 ARP Steps for Same Network

Step	Action
1	Sender knows receiver name; DNS returns IP2
2	IP software checks: destination is on **same network**
3	Needs Ethernet address E2 to send frame
4	Host 1 **broadcasts ARP**: "Who owns IP 192.32.65.5?"
5	All machines check their IP address
6	Only Host 2 responds with Ethernet address E2
7	IP software builds frame to E2, puts IP packet in payload
8	Host 2 receives, extracts IP packet, passes to IP software

ARP Optimizations

Table 5-45 ARP Optimization Techniques

Technique	Mechanism	Benefit
Caching	Cache ARP result after first use	Eliminates second broadcast
Include sender mapping	Sender includes its (IP, Ethernet) mapping	Receiver caches it too
Timeout	Cache entries time out after minutes	Allows mappings to change
Gratuitous ARP	Broadcast mapping when configured	Updates everyone's cache
Duplicate detection	If response to gratuitous ARP arrives	Two machines have same IP - error

ARP Operation: Inter-Network (Host 1 → Host 4)

Table 5-46 ARP Steps for Different Networks

Step	Action
1	IP software sees IP4 = 192.32.63.8 is **NOT** on CS network
2	Off-network traffic sent to **default gateway** (192.32.65.1)
3	Host 1 ARP broadcasts for 192.32.65.1, learns E3
4	Host 1 sends frame to E3 with IP packet still destined for IP4
5	Router receives; sees destination is on EE network
6	Router ARPs for 192.32.63.8 on EE network, learns E6
7	Router sends frame to E6; IP packet still destined for IP4

> **Key Observation**
>
> **Ethernet addresses change with each network hop; IP addresses remain constant** across the entire path.

Proxy ARP

> **Proxy ARP**
>
> Router answers ARPs on one network for a host on another network, giving its own Ethernet address. Used when a host doesn't know the destination is on a different network.

DHCP - The Dynamic Host Configuration Protocol

> **The Configuration Problem**
>
> Manual IP configuration is tedious and error-prone for thousands of machines.

DHCP Process

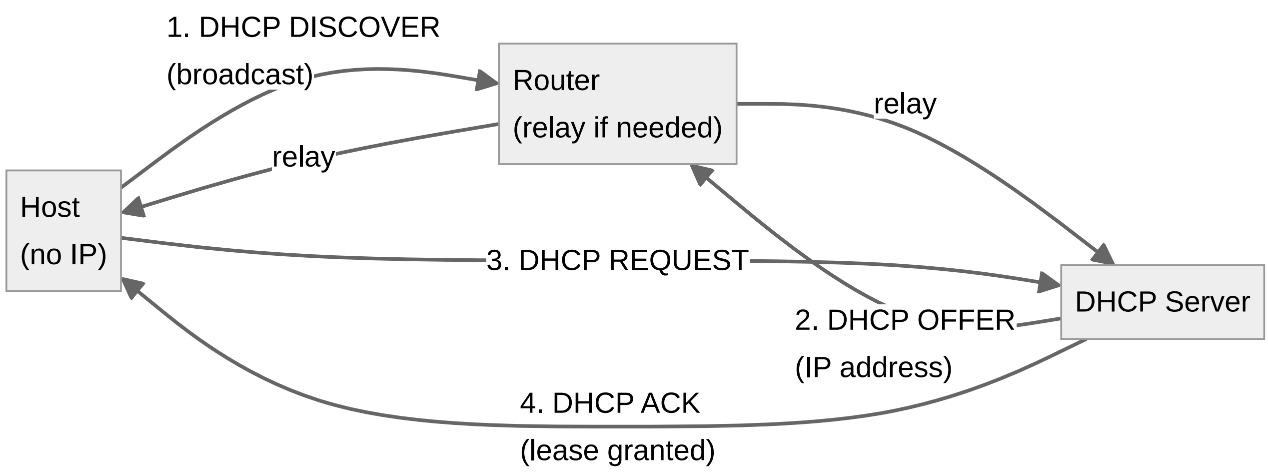


Table 5-47 DHCP Steps

Step	Action
1. DHCP DISCOVER	Computer broadcasts request for IP address
2. DHCP OFFER	Server allocates free IP, sends to host
3. DHCP REQUEST	Host requests offered IP
4. DHCP ACK	Server confirms; lease granted

Leasing

Table 5-48 DHCP Leasing

Aspect	Detail
Problem	If host leaves without returning IP, address is permanently lost
Solution	**Leasing** - IP assignment for fixed period
Renewal	Host must request renewal before lease expires
Configured params	Network mask, default gateway, DNS server, time server

> **DHCP Predecessors**
>
> DHCP largely replaced **RARP** and **BOOTP** (more limited functionality).

Exam Tip: Key Takeaways for 5.6

1. **Internet design:** 10 core principles emphasizing simplicity, modularity, scalability.
2. **Internet architecture:** Collection of interconnected autonomous systems (Tier 1, ISPs, regional, edge).
3. **IPv4:** Best-effort, connectionless; 20-byte header + options; fragmentation via ID/DF/MF/offset.
4. **IP addressing:** Hierarchical 32-bit with CIDR notation; subnetting splits prefixes internally.
5. **CIDR:** Route aggregation reduces core router tables; longest matching prefix for overlapping routes.
6. **Classful addressing (pre-1993):** A/B/C/D/E classes led to "three bears" address waste problem.
7. **NAT:** Quick fix for IPv4 exhaustion using private ranges + port translation; violates IP architecture.
8. **IPv6:** 128-bit addresses, 40-byte fixed header (7 fields), extension headers, no header checksum.
9. **ICMP:** Error reporting (destination unreachable, time exceeded); ping (echo), traceroute (TTL).
10. **ARP:** Resolves IP to Ethernet via broadcast; DHCP automates host configuration via leasing.

Exam Questions & Answers

How to Use This Section

This section contains both **theoretical** and **numerical** questions commonly asked in exams. Practice answering them before checking the solutions. Numerical questions are marked with an orange bar.

Section A: Theoretical Questions

Q1. What is store-and-forward packet switching? Explain with a diagram.

Answer: Store-and-forward packet switching is the fundamental mechanism where a packet is **fully received and stored** at each router before being **forwarded** to the next hop. The process: (1) Host sends packet to nearest router; (2) Router stores the packet until fully arrived and verified (checksum); (3) Packet is forwarded to the next router; (4) Process repeats until destination. This introduces a store-and-forward delay at each hop equal to the packet size divided by the link bandwidth.

Q2. Compare connection-oriented and connectionless services in the network layer. List two examples of each.

Answer:

Feature	Connectionless	Connection-Oriented
Setup	Not needed	Required before data transfer
Addressing	Full address per packet	Short VC number
Routing	Each packet independent	Fixed route for all packets
State	No connection state in routers	Router table space per VC
QoS	Difficult	Easy with resource reservation
Examples	IP (Internet Protocol)	MPLS, ATM, X.25

Q3. What is the end-to-end argument? How does it relate to the Internet's design philosophy?

Answer: The **end-to-end argument** (Saltzer et al., 1984) states that functions placed at lower layers of a system may be redundant or of little value when compared to the cost of providing them there. The Internet community applied this by keeping the network layer simple (connectionless, no error control) and pushing reliability to the end hosts (transport layer). This makes routers simpler and faster, and hosts can implement whatever error control they need.

Q4. Explain the optimality principle in routing. What is a sink tree?

Answer: The **optimality principle** (Bellman, 1957) states: If router J is on the optimal path from I to K, then the optimal path from J to K also falls along the same route. Proof by contradiction: if a better path from J to K existed, it could be combined with I-to-J to create a better I-to-K path. A **sink tree** is the set of optimal routes from all sources to a given destination, forming a tree rooted at that destination. It has no loops and guarantees finite delivery.

Q5. Differentiate between routing and forwarding.

Answer:

- **Routing:** The process of making decisions about which routes to use. It's the **planning phase** - computing paths and populating routing tables. Happens in background.
- **Forwarding:** The process of handling an arriving packet - looking up the outgoing line in the routing table and sending it on its way. It's the **action phase** - happens for every packet.

A router has two processes: one for forwarding packets and one for running the routing algorithm to update tables.

Q6. What are the desirable properties of a routing algorithm? Explain any two in detail.

Answer: Desirable properties: correctness, simplicity, robustness, stability, fairness, efficiency.

- **Robustness:** Must cope with hardware/software failures, topology changes, and line failures without requiring network-wide reboots. The algorithm should adapt to changes automatically.
- **Fairness vs. Efficiency:** Fairness means treating connections equitably. Efficiency means maximizing total throughput. These are often in conflict - maximizing total flow might

require shutting off some connections entirely, which fairness would not accept. A compromise is necessary.

Q7. Compare static and dynamic routing algorithms.

Answer:

Feature	Static (Nonadaptive)	Dynamic (Adaptive)
Information used	No current measurements	Current topology and traffic
Computation	In advance, offline	Continuously during operation
Response to failures	None	Adapts to new topology
Complexity	Simple	More complex
Best for	Stable, small networks	Large, changing networks

Q8. What is the count-to-infinity problem in distance vector routing? Why does it occur?

Answer: The **count-to-infinity problem** is the slow convergence of distance vector routing when bad news (link failure) occurs. Good news spreads at one hop per exchange, but bad news propagates slowly because routers gradually increment their distance estimates toward infinity. The core problem: **when X tells Y it has a path somewhere, Y has no way of knowing if it itself is on that path**. So Y may choose a path that actually goes through itself. Split horizon with poisoned reverse is an attempted solution but doesn't work well in practice.

Q9. Explain Dijkstra's shortest path algorithm. What are tentative and permanent labels?

Answer: Dijkstra's algorithm (1959) finds shortest paths from a source to all destinations using non-negative edge weights. Each node has a label (distance from source):

- **Tentative label:** Initial state, may be improved as algorithm proceeds. Initially set to infinity.
- **Permanent label:** Confirmed shortest path from source; never changed.

Steps: (1) Make source permanent with distance 0; (2) Examine neighbors of newest permanent node, update tentative labels; (3) Select tentative node with smallest label, make permanent; (4) Repeat until all nodes permanent.

Q10. List the 10 design principles of the Internet. Which is the most important?

Answer: Top 10 (RFC 1958): 1. Make sure it works; 2. Keep it simple; 3. Make clear choices; 4. Exploit modularity; 5. Expect heterogeneity; 6. Avoid static options; 7. Good design need not be perfect; 8. Be strict sending, tolerant receiving; 9. Think scalability; 10. Consider performance and cost. The most important is **"Make sure it works"** - don't finalize until multiple prototypes have successfully communicated.

Q11. Describe the IPv4 header format with a diagram. What is the purpose of the TTL field?

Answer: The IPv4 header has a 20-byte fixed part + variable options. Key fields: Version (4 bits), IHL (4 bits), Differentiated Services (8 bits), Total Length (16 bits), Identification (16 bits), Flags + Fragment Offset (16 bits), TTL (8 bits), Protocol (8 bits), Header Checksum (16 bits), Source Address (32 bits), Destination Address (32 bits), Options (variable). **TTL (Time to Live):** Originally meant time in seconds, now counts hops. Decremented at each router; packet discarded when reaches 0. Prevents packets from wandering forever if routing tables are corrupted. Maximum: 255 hops.

Q12. What is CIDR? How does it help reduce routing table size?

Answer: **CIDR (Classless InterDomain Routing)** allows variable-length network prefixes, eliminating fixed class boundaries (A/B/C). It enables **route aggregation**: combining multiple small prefixes into a single larger prefix (supernet). For example, three university prefixes (/21, /22, /20) can be advertised as one /19 aggregate to distant routers. This dramatically reduces core routing table entries. Routers use **longest matching prefix** to handle overlapping prefixes.

Q13. Explain subnetting with an example. Why is it needed?

Answer: **Subnetting** splits a block of addresses into multiple internal networks while appearing as one network externally. **Need:** A university with a /16 for CS later needs to add EE and Art departments. Instead of requesting new blocks (expensive, wasteful), they subnet internally. **Example:** 128.208.0.0/16 can be split into CS (/17 = 32,768 addresses), EE (/18 = 16,384), Art (/19 = 8,192), with 8,192 unallocated. The main router uses subnet masks to forward packets to the correct department.

Q14. What is NAT? How does it work and what are its main objections?

Answer: **NAT (Network Address Translation, RFC 3022)** maps private internal addresses to a public IP address for Internet traffic. It uses TCP/UDP port numbers to distinguish connections from different internal hosts. **Key objections:** (1) Violates IP architecture - thousands of machines share one IP; (2) Breaks end-to-end connectivity - incoming connections need prior outgoing traffic; (3) Makes Internet connection-oriented - NAT box maintains state; (4) Violates protocol layering - inspects layer 4 headers from layer 3; (5) Breaks non-TCP/UDP protocols; (6) Breaks applications with embedded IP addresses (FTP, H.323); (7) Limited by 16-bit port space (~61,440 machines per IP).

Q15. Compare IPv4 and IPv6 headers. What fields were removed and why?

Answer:

Feature	IPv4	IPv6
Header size	20-60 bytes (variable)	40 bytes (fixed)
Address size	32 bits	128 bits
Fields	13 fields	7 fields
Checksum	Present (recomputed each hop)	Removed (other layers handle it)
Fragmentation	Routers can fragment	Only source can fragment
Options	In header (variable)	Extension headers (separate)
Flow label	Not present	20-bit flow label for QoS

Removed: IHL (fixed length), Protocol (replaced by Next Header), Fragmentation fields (different approach), Header Checksum (performance not worth value).

Q16. What are ICMP and ARP? Explain their functions with examples.

Answer: **ICMP (Internet Control Message Protocol):** Reports unexpected events during packet processing. Encapsulated in IP packets. Examples: Destination Unreachable (can't deliver), Time Exceeded (TTL=0), Echo Request/Reply (used by ping). **ARP (Address Resolution Protocol):** Resolves IP addresses to Ethernet (MAC) addresses. Host broadcasts "Who has IP X?" and the target responds with its MAC address. Result is cached. Used for intra-network communication. For inter-network, ARP is used to find the router's MAC address.

Q17. How does DHCP work? What is a lease?

Answer: **DHCP (Dynamic Host Configuration Protocol)** automates host IP configuration. Process: (1) Host broadcasts DHCP DISCOVER; (2) Server responds with DHCP OFFER (IP address); (3) Host sends DHCP REQUEST; (4) Server sends DHCP ACK. A **lease** is a time-limited IP assignment. Hosts must renew before expiry or lose the address. DHCP also provides network mask, default gateway, and DNS server addresses.

Section B: Numerical Problems

N1. A university has the IP block 192.168.0.0/16. They need to create subnets for 4 departments: Engineering (500 hosts), Science (200 hosts), Arts (100 hosts), and Admin (50 hosts). Design an efficient subnetting scheme.

Answer:

- **Engineering (500 hosts):** Need 9 host bits (2^9 - 2 = 510). Prefix: **192.168.0.0/23**, Mask: 255.255.254.0
- **Science (200 hosts):** Need 8 host bits (2^8 - 2 = 254). Prefix: **192.168.2.0/24**, Mask: 255.255.255.0
- **Arts (100 hosts):** Need 7 host bits (2^7 - 2 = 126). Prefix: **192.168.3.0/25**, Mask: 255.255.255.128

- **Admin (50 hosts):** Need 6 host bits (2^6 - 2 = 62). Prefix: **192.168.3.128/26**, Mask: 255.255.255.192

N2. Given IP address 172.16.5.130/22, find: (a) Network address, (b) Broadcast address, (c) First usable host, (d) Last usable host, (e) Number of usable hosts.

Answer:

- /22 means 22 network bits, 10 host bits. Subnet mask: 255.255.252.0
- Block size in 3rd octet: 256 - 252 = 4. Network blocks: 172.16.0.0, 172.16.4.0, 172.16.8.0...
- 172.16.5.130 falls in **172.16.4.0/22**
- (a) **Network:** 172.16.4.0
- (b) **Broadcast:** 172.16.7.255 (next network - 1)
- (c) **First usable:** 172.16.4.1
- (d) **Last usable:** 172.16.7.254
- (e) **Usable hosts:** 2^{10} - 2 = 1022

N3. In distance vector routing, router J has neighbors A (delay 8), I (delay 10), H (delay 12), K (delay 6). Neighbor A advertises: G=18, B=12, C=25. Neighbor H advertises: G=6, B=31, C=19. Calculate J's delay to G and B, and which neighbor to use.

Answer: **To G:**

- Via A: 18 + 8 = 26
- Via I: (not given, assume no route)
- Via H: 6 + 12 = **18 (Best)**
- Via K: (not given, assume no route)

J's route to G: Delay = 18, via H
To B:

- Via A: 12 + 8 = **20 (Best)**
- Via H: 31 + 12 = 43

J's route to B: Delay = 20, via A

N4. Using Dijkstra's algorithm, find the shortest path from A to all other nodes in the following network: A-B(2), A-C(4), B-C(1), B-D(7), C-D(3), C-E(5), D-E(1), D-F(2), E-F(3).

Answer:

Step	Permanent Node	B	C	D	E	F
0	A(0)	∞	∞	∞	∞	∞
1	A(0)	2(A)	4(A)	∞	∞	∞
2	B(2)	2(A)	3(B)	9(B)	∞	∞
3	C(3)	2(A)	3(B)	6(C)	8(C)	∞
4	D(6)	2(A)	3(B)	6(C)	7(D)	8(D)
5	E(7)	2(A)	3(B)	6(C)	7(D)	8(D)
6	F(8)	2(A)	3(B)	6(C)	7(D)	8(D)

Shortest paths from A: B=2 (A-B), C=3 (A-B-C), D=6 (A-B-C-D), E=7 (A-B-C-D-E), F=8 (A-B-C-D-F)

N5. An IPv4 packet has Total Length = 4000 bytes, Identification = 0x1A2B, DF=0, MF=0, Fragment Offset = 0. It needs to traverse a link with MTU = 1500 bytes. Show all fragments with their header field values.

Answer:

- IP header = 20 bytes. Max data per fragment = 1500 - 20 = 1480 bytes.
- Fragment offset unit = 8 bytes. 1480/8 = 185 (exact division).
- Original data = 4000 - 20 = 3980 bytes.

Fragment	Data Bytes	Offset	MF	Total Length
1	1480 (bytes 0-1479)	0	1	1500
2	1480 (bytes 1480-2959)	185	1	1500
3	1020 (bytes 2960-3979)	370	0	1040

All fragments: Identification = 0x1A2B, DF = 0.

N6. A router has these entries: 192.168.0.0/16 (port 1), 192.168.1.0/24 (port 2), 192.168.2.0/24 (port 3), 0.0.0.0/0 (port 4). Determine the outgoing port for: (a) 192.168.1.5, (b) 192.168.3.10, (c) 192.168.0.50, (d) 10.0.0.1.

Answer: Using **longest matching prefix**:

- (a) 192.168.1.5: Matches /16 (port 1) and /24 (port 2). Longest: **/24 → Port 2**
- (b) 192.168.3.10: Matches /16 (port 1) only. **Port 1**
- (c) 192.168.0.50: Matches /16 (port 1) only. **Port 1**
- (d) 10.0.0.1: No specific match. **Default → Port 4**

N7. A network has 5 routers in a line: A-B-C-D-E. Initially A is down. Show the count-to-infinity problem if the A-B link fails after all routers have converged. Assume infinity = 16.

Answer:

Initial converged state (distances to A): B=1, C=2, D=3, E=4

After A-B link fails:

Exchange	B	C	D	E	Explanation
Initial	1	2	3	4	Normal state
1	3	2	3	4	B hears C has path=2, so B via C = 3
2	3	4	3	4	C hears B=3 or D=3, picks B, updates to 4
3	5	4	5	4	D updates to 5
4	5	6	5	6	E updates to 6
5	7	6	7	6	Continuing...
...	...	...	...	...	Gradually approaches 16 (infinity)

This is the count-to-infinity problem. It takes many exchanges to converge to infinity.

N8. A company has 2000 employees and receives the block 203.10.20.0/21. They need subnets for: HQ (800 hosts), Branch1 (400 hosts), Branch2 (200 hosts), Branch3 (100 hosts), and a server farm (50 hosts). Design the subnetting.

Answer:

- /21 = 2048 addresses total. Available: 203.10.16.0 - 203.10.23.255
- **HQ (800):** Need 10 host bits (2^{10}-2=1022). **203.10.16.0/22** (203.10.16.0-203.10.19.255)
- **Branch1 (400):** Need 9 host bits (2^{9}-2=510). **203.10.20.0/23** (203.10.20.0-203.10.21.255)
- **Branch2 (200):** Need 8 host bits (2^{8}-2=254). **203.10.22.0/24** (203.10.22.0-203.10.22.255)
- **Branch3 (100):** Need 7 host bits (2^{7}-2=126). **203.10.23.0/25** (203.10.23.0-203.10.23.127)
- **Server farm (50):** Need 6 host bits (2^{6}-2=62). **203.10.23.128/26** (203.10.23.128-203.10.23.191)

N9. In a classful addressing system, determine the class, default mask, network address, and broadcast address for: (a) 10.5.3.2, (b) 172.16.20.5, (c) 192.168.1.1, (d) 224.0.0.1.

Answer:

Address	Class	Default Mask	Network	Broadcast
10.5.3.2	A (1-126)	255.0.0.0 (/8)	10.0.0.0	10.255.255.255
172.16.20.5	B (128-191)	255.255.0.0 (/16)	172.16.0.0	172.16.255.255
192.168.1.1	C (192-223)	255.255.255.0 (/24)	192.168.1.0	192.168.1.255
224.0.0.1	D (224-239)	N/A	N/A	Multicast

N10. A NAT box translates internal network 192.168.1.0/24 to public IP 203.0.113.5. Two hosts (192.168.1.10 and 192.168.1.20) both want to connect to a web server at 198.51.100.10:80. Host 1 uses source port 5000, Host 2 uses source port 5000. Show the NAT translation table after both connections are established.

Answer:

NAT Port	Internal IP	Internal Port	Remote IP	Remote Port
40001	192.168.1.10	5000	198.51.100.10	80
40002	192.168.1.20	5000	198.51.100.10	80

- **Outgoing Host 1:** Source 192.168.1.10:5000 → 203.0.113.5:40001
- **Outgoing Host 2:** Source 192.168.1.20:5000 → 203.0.113.5:40002
- The NAT box **must replace the source port** because both hosts use the same port 5000. The NAT port (40001, 40002) uniquely identifies each connection.
- **Return traffic:** Packets to 203.0.113.5:40001 forwarded to 192.168.1.10:5000; packets to 203.0.113.5:40002 forwarded to 192.168.1.20:5000.

N11. For the IPv6 address 2001:0DB8:0000:0000:0000:FF00:0042:8329, write: (a) The compressed form, (b) The :: notation form.

Answer:

- (a) **Compressed (omit leading zeros):** 2001:DB8:0:0:0:FF00:42:8329
- (b) **:: notation (replace longest zero run): 2001:DB8::FF00:42:8329**
- Note: Only one :: allowed per address to avoid ambiguity.

N12. Using Figure 5-9 from the notes, router J receives vectors from neighbors A (delay 8), I (delay 10), H (delay 12), K (delay 6). Complete J's new routing table for destinations B, C, D, E, F, G, L using the neighbor data below.

Answer:

Dest	From A (delay 8)	From I (delay 10)	From H (delay 12)	From K (delay 6)	J's Delay	J's Line
B	12	36	31	28	20 (12+8)	A
C	25	18	19	36	28 (18+10)	I
D	40	27	8	24	20 (8+12)	H
E	14	7	30	22	17 (7+10)	I
F	23	20	19	40	30 (20+10)	I
G	18	31	6	31	18 (6+12)	H
L	29	33	9	9	15 (9+6)	K

Section C: Quick Reference Tables

Table R-1 Quick Reference: Key Protocols

Protocol	Layer	Purpose	Key Feature
IP (IPv4/IPv6)	Network	Best-effort packet delivery	Connectionless, 32/128-bit addresses
ICMP	Network	Error reporting, testing	Ping, traceroute, TTL exceeded
ARP	Network	IP to MAC resolution	Broadcast query, cached replies
DHCP	Application	Auto IP configuration	Lease-based address assignment
NAT	Network edge	Private to public translation	Port-based multiplexing
MPLS	Network (ISP)	Fast packet forwarding	Label switching, connection-oriented

Table R-2 Quick Reference: Important RFCs

RFC	Topic
RFC 791	IPv4 Protocol
RFC 1918	Private Address Space
RFC 1958	Architectural Principles of the Internet
RFC 2131	DHCP
RFC 2460	IPv6 Specification
RFC 3022	Traditional NAT

Table R-3 Quick Reference: Common Port Numbers

Protocol	Port	Purpose
DHCP	67/68	Dynamic host configuration
DNS	53	Domain name resolution
HTTP	80	Web traffic
HTTPS	443	Secure web traffic
SSH	22	Secure shell

Final Exam Checklist

- Store-and-forward packet switching mechanism
- Connection-oriented vs. connectionless services and their trade-offs
- Routing vs. forwarding distinction
- Dijkstra's algorithm - be able to execute step-by-step
- Distance vector routing and count-to-infinity problem
- Sink tree concept and optimality principle
- IPv4 header fields - memorize key fields and their purposes
- IP addressing - CIDR notation, subnetting calculations
- Longest matching prefix routing
- NAT operation and objections
- IPv6 header - what changed from IPv4
- ICMP message types and their uses
- ARP process - intra-network and inter-network
- DHCP four-step process and leasing